

Test Execution Report

IBVAP — Intelligent Border Video Analytics Platform · Problem Statement 26187, Ministry of Home Affairs (SSB)

247	100%	0	82.1s	76.1%
tests executed	pass rate	failures / errors	wall clock	core coverage

Result

The full suite passed. **247 of 247 tests** executed with zero failures, zero errors and zero skips, in 82.1 seconds. Static analysis (ruff, 8 rule families across 110 files) reported no findings.

The suite is split between fast unit tests that isolate one behaviour each, and integration tests that exercise the real pipeline end to end — a live synthetic camera through detection, tracking, analytics, evidence capture and the HTTP API. No component is mocked in the integration path except the model artefacts, which are deliberately absent so the platform's degraded-mode behaviour is what gets tested.

Environment

Field	Value
Platform	Linux 6.18.44 · x86-64 · 4 cores · 15 GB RAM
Python	3.11.15
Key libraries	opencv 5.0.0 · onnxruntime 1.29.0 · numpy 2.4.6 · FastAPI 0.141 · SQLAlchemy 2.0.52 · PyQt6 6.11
Runner	pytest 8.x with pytest-asyncio, pytest-cov
Commit	d8dbabe on branch tedo
Executed	26 August 2026, 13:25

Coverage by subsystem

Percentages are statement coverage from the same run. The desktop console is listed separately: it is GUI code, verified by driving the real application rather than by unit test, and including it in a single headline figure would misrepresent both numbers.

Subsystem	Scope	Stmts	Cov.	
core	Domain types, config, geometry, time windows	764	92%	
telemetry	Prometheus instrumentation	89	88%	
analytics	Rules engine and event suppression gate	580	83%	
ingest	Video sources, reader, bounded queue	354	82%	
events	Annotation, evidence, canonical payload	409	80%	
mlops	Registry, export, benchmark, evaluation, drift	679	79%	
vision	Detector, tracker, ANPR, face, preprocessing	1254	73%	
api	FastAPI app, auth/RBAC, routers	1188	71%	
pipeline	Model bundle, camera worker, supervisor	436	70%	
storage	ORM, async engine, repositories	460	70%	
integrations	Webhook, MQTT, syslog, dispatcher	403	60%	
Core platform	all of the above	6616	76%	

Desktop console (PyQt6): 1,125 statements, 0% automated. Verified by launching the real application against a live four-camera node — video wall streaming, 200 alerts listed, evidence integrity verified, and zone geometry drawn and confirmed persisted by re-reading the camera configuration from the node.

What the tests verify

Grouped by the subsystem under test. Counts are individual test cases; parametrised cases are counted once per parameter set.

Geometry & virtual fencing — 17 tests

- Polygon containment, including concave shapes where convex-hull logic would be wrong
- Zones defined in normalised coordinates cover the same scene at 1080p, 720p and 360p
- Movement past the *extension* of a tripwire does not fire — the classic false-alarm bug
- Configured direction matches the arrow the console draws, for three wire orientations

Object tracking — 17 tests

- Identity survives at 5, 10, 20, 35 and 50 px/frame — the Kalman filter learns velocity
- A track recovers its identity after a five-frame occlusion, and expires past its age budget
- Low-confidence detections continue a track (the ByteTrack property that keeps IR-flickering subjects alive)
- A vehicle passing behind a sentry cannot steal their track id; crossing paths keep their identities
- Object class is majority-voted over the track, not taken from the latest frame

ANPR — 29 tests

- Eight valid Indian plate formats pass through unchanged, including Bharat series
- Four OCR confusion patterns are repaired: MHI2A8I234 → MH12AB1234
- An unrecognised state code is repaired only when a real RTO code exists; otherwise marked invalid
- Confidence is penalised per correction; a low-confidence read is never actionable
- CTC decoding collapses repeats and strips blanks
- Watchlist entries are normalised on entry, so a mistyped plate cannot sit inert

Analytics rules — 20 tests

- Intrusion requires confirmation frames — single-frame detector jitter does not alert
- Zone membership is tested at the object's feet, not its centroid
- Livestock is suppressed by class — the dominant false-alarm source on rural fences
- Loitering distinguishes dwelling from traversing; abandoned object requires no owner nearby
- Night rules respect schedule windows that wrap midnight
- Camera tamper detects a covered lens while the stream stays up, and does not fire on a healthy scene
- The gate deduplicates, applies cooldown, rate-limits, and prunes its own memory

Vision pipeline — 26 tests

- YOLOv5, YOLOv8 and YOLO26 output layouts all decode correctly
- An end-to-end (YOLO26) head keeps two heavily overlapping people — NMS is correctly skipped
- The letterbox inverse is exact, and clamps boxes predicted inside the padding
- Class-aware NMS keeps a motorcyclist's person and motorcycle boxes
- The classical fallback suppresses output during background warm-up and caps its confidence
- Face matching rejects an ambiguous probe on margin, not just threshold

Evidence & chain of custody — 14 tests

- Snapshots are written, hashed and recorded in a manifest
- Altering a stored snapshot is detected
- Deleting a manifest breaks the day's chain and is reported
- Retention prunes by both age and total size

Security — 19 tests

- Bcrypt hashing, with SHA-256 pre-hash so >72-byte passphrases stay distinct
- Role hierarchy; an unrecognised role fails closed to viewer
- A refresh token is rejected where an access token is required
- Revocation works; logout is keyed on username and source together

MLOps — 31 tests

- mAP with all-point interpolation: perfect, offset, duplicate and missed detections
- ANPR scored through the grammar stage, including reads it would corrupt
- Registry refuses to mutate a published version; a checksum mismatch is fatal
- Drift separates steady state from illuminator failure and camera re-aim

C2 integrations — 18 tests

- HMAC signatures verify, and a tampered body or stale timestamp is rejected
- CEF output is well-formed and stays within the standard cs1–cs6 slots
- Severity and event-type filtering per sink; retry backoff schedule

Configuration — 24 tests

- A rule referencing a non-existent zone is rejected at load
- Zone points outside [0,1] are rejected; duplicate camera ids are caught
- Environment variables override YAML; midnight-wrapping windows partition the day

Integration (end-to-end) — 32 tests

- A live synthetic camera produces intrusion and line-crossing alerts through the real pipeline
- Frames are sampled from source rate to the analytics rate
- An unreachable camera does not stop the others from working
- Cameras can be added and removed at runtime
- Full API surface: auth, RBAC denial, events with evidence, media query-token scoping

Slowest tests

All of the slowest cases are integration tests that start a real node and wait for live video to produce alerts. The duration is dominated by deliberate wall-clock waits, not by computation.

Duration	Test
6.04 s	test_pipeline · frames are sampled to the target rate
5.04 s	test_pipeline · unreachable camera does not stop the others
4.66 s	test_api · events appear and carry evidence
4.03 s	test_pipeline · produces alerts from a live camera
3.03 s	test_pipeline · reports health and degradation
2.70 s	test_api · media endpoints accept a query token
2.15 s	test_api · readiness reflects camera state
2.03 s	test_pipeline · cameras can be added and removed at runtime

Measured throughput

Captured on the same machine during this run, using the classical fallback detector. Camera capacity is sized on p95 latency with 30% headroom — a node planned to its median capacity has no margin for the slow frames, which arrive exactly when a scene becomes busy.

Stage	p50	p95	p99	Throughput	Cameras @ 8 fps
Detection	4.90 ms	10.28 ms	14.12 ms	204 fps	8.5
Tracking	0.89 ms	1.28 ms	1.41 ms	1118 fps	68.3

Detection dominates cost by roughly 5:1. Raising `detect_interval` is therefore the effective lever when a node is oversubscribed; shortening track trails is not.

What this run does not establish

Stated plainly, because a green suite invites more confidence than it earns.

Not covered	Why, and what it means
Neural model accuracy	No model artefacts exist in the repository — they are registry-managed, not committed. Every test ran against the classical fallback detector. The mAP, ANPR and face evaluators are themselves tested, but against synthetic data, so no statement is made here about real-world detection accuracy.
Desktop GUI (1,125 stmts)	Verified manually by driving the real application against a live node, not by automated test. A regression in Qt widget behaviour would not be caught by this suite.
Container image	The Dockerfile and Compose stack are written and the Compose schema validates, but the image was never built here — the sandbox gateway refuses Docker Hub CDN requests. Build it before relying on it.
Real RTSP cameras	All video came from the in-process synthetic source. Reconnection and backoff were tested against a refused TCP endpoint, which exercises the failure path but not a real NVR's behaviour under load.
Sustained load and soak	The longest test runs six seconds. Memory growth, evidence disk pressure and outbox behaviour over days are not measured here.
Multi-node / central tier	Only single-node operation was exercised. PostgreSQL and the Kubernetes manifests are untested.

Defects found and fixed during development

Recorded because they show what the suite was built to catch. Each was found by running the code, not by inspection.

Defect	Consequence had it shipped
Kalman filter seeded with over-tight velocity covariance	Tracks fragmented every few frames on any fast mover; no dwell or crossing rule could fire on them.
<code>selected_zones()</code> defaulted to all zones	Rules with no zone configured became silently zone-restricted. Rapid-movement, abandoned-object and night-movement were inert.
<code>ObjectClass.coerce("animal")</code> returned UNKNOWN	The livestock-suppression setting — the highest-value tuning knob in this domain — did nothing.
Fancy-indexed clip in the letterbox inverse	Boxes predicted inside the padding kept negative coordinates, placing detections outside the frame.
Blocking <code>stat()</code> in an async handler	Stalled the event loop, delaying every other request the node was serving.
passlib depends on crypt, removed in Python 3.13	The platform would fail to start on 3.13. Replaced with direct bcrypt.
Rule registry never populated on import	A camera would load with all analytics silently disabled.
<code>[hidden]</code> overridden by <code>display:grid</code>	The browser login screen never hid after a successful sign-in.
Zone editor used POST where the API expects PUT	Drawing a fence appeared to work; saving silently did nothing.

Reproduce this report

```
pytest tests/ -q --cov=ibvap --durations=15
ruff check src tests
```

Fast subset (unit tests only, ~5 s): `pytest tests/unit -q`